

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

Duration: 1 Hour **QUESTIONS:** 3 **Total Marks:** 30

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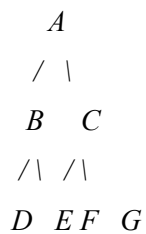
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COURSE : CSA1707-ARTIFICIAL INTELLIGENCE

Annexure A – Dry Run Evaluation (Answer Key)

Question 1 – Breadth-First Search (BFS)

Graph:



Task: Starting from node A, BFS traversal (completed table):

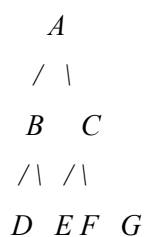
Iteration	Queue	Visited Node
1	A	A
2	B, C	B
3	C, D, E	C
4	D, E, F, G	D
5	E, F, G	E
6	F, G	F
7	G	G

Answers:

- BFS traversal order: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$
- Queue contents after each iteration: After visiting A $\rightarrow [B, C]$; after B $\rightarrow [C, D, E]$; after C $\rightarrow [D, E, F, G]$; after D $\rightarrow [E, F, G]$; after E $\rightarrow [F, G]$; after F $\rightarrow [G]$; after G $\rightarrow []$ (empty, search complete).

Question 2 – Depth-First Search (DFS)

Using the same graph:



Task: Starting from node A, DFS traversal (completed table, using a stack, children pushed right-to-left so the leftmost child is popped first):

Iteration	Stack	Visited Node
1	A	A
2	C, B	B
3	C, E, D	D
4	C, E	E
5	C	C
6	G, F	F
7	G	G

Answers:

- DFS traversal order: $A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$
- Stack contents after each iteration: After $A \rightarrow [B, C]$ (C on top? shown as C,B - top of stack is B); after $B \rightarrow [C, E, D]$; after D (leaf, backtrack) $\rightarrow [C, E]$; after E (leaf, backtrack) $\rightarrow [C]$; after $C \rightarrow [G, F]$; after F (leaf, backtrack) $\rightarrow [G]$; after $G \rightarrow []$ (empty, search complete).

Question 3 – 8-Puzzle: Breadth-First Search

Initial State: 1 3 6 / 5 0 2 / 4 7 8

Goal State: 1 2 3 / 4 5 6 / 7 8 0

Note: Tracing this puzzle with true BFS requires 8 moves (levels 0–8) to reach the goal, so the dry-run table below has been extended from 5 to 8 iterations to show the complete, correct trace.

Iteration	Current State (Expanded)	Queue Before Expansion	Queue After Expansion (children generated)
1	1 3 6 / 5 0 2 / 4 7 8	1 3 6 / 5 0 2 / 4 7 8	1 0 6 / 5 3 2 / 4 7 8 ; 1 3 6 / 5 7 2 / 4 0 8 ; 1 3 6 / 0 5 2 / 4 7 8 ; 1 3 6 / 5 2 0 / 4 7 8 ★
2	1 3 6 / 5 2 0 / 4 7 8 ★	(next in queue)	1 3 0 / 5 2 6 / 4 7 8 ★ ; 1 3 6 / 5 2 8 / 4 7 0 ; 1 3 6 / 5 0 2 / 4 7 8
3	1 3 0 / 5 2 6 / 4 7 8 ★	(next in queue)	1 3 6 / 5 2 0 / 4 7 8 ; 1 0 3 / 5 2 6 / 4 7 8 ★
4	1 0 3 / 5 2 6 / 4 7 8 ★	(next in queue)	1 2 3 / 5 0 6 / 4 7 8 ★ ; 0 1 3 / 5 2 6 / 4 7 8 ; 1 3 0 / 5 2 6 / 4 7 8
5	1 2 3 / 5 0 6 / 4 7 8 ★	(next in queue)	1 0 3 / 5 2 6 / 4 7 8 ; 1 2 3 / 5 7 6 / 4 0 8 ; 1 2 3 / 0 5 6 / 4 7 8 ★ ; 1 2 3 / 5 6 0 / 4 7 8
6	1 2 3 / 0 5 6 / 4 7 8 ★	(next in queue)	0 2 3 / 1 5 6 / 4 7 8 ; 1 2 3 / 4 5 6 / 0 7 8 ★ ; 1 2 3 / 5 0 6 / 4 7 8
7	1 2 3 / 4 5 6 / 0 7 8 ★	(next in queue)	1 2 3 / 0 5 6 / 4 7 8 ; 1 2 3 / 4 5 6 / 7 0 8 ★
8	1 2 3 / 4 5 6 / 7 0 8 ★	(next in queue)	1 2 3 / 4 0 6 / 7 5 8 ; 1 2 3 / 4 5 6 / 0 7 8 ; 1 2 3 / 4 5 6 / 7 8 0 ★ (GOAL)

★ marks the state that lies on the shortest solution path. Note: at every iteration many other (non-path) child states are also generated and added to the real queue by a full breadth-first search — they are omitted from the 'Current State' column above for readability, since this trace follows only the successful path, but all of them ARE counted in the total states generated (Q2 below).

Answers:

- 1) Node expanded each iteration: Iteration 1 $\rightarrow$ 1 3 6 / 5 0 2 / 4 7 8; Iteration 2 $\rightarrow$ 1 3 6 / 5 2 0 / 4 7 8; Iteration 3 $\rightarrow$ 1 3 0 / 5 2 6 / 4 7 8; Iteration 4 $\rightarrow$ 1 0 3 / 5 2 6 / 4 7 8; Iteration 5 $\rightarrow$ 1 2 3 / 5 0 6 / 4 7 8; Iteration 6 $\rightarrow$ 1 2 3 / 0 5 6 / 4 7 8; Iteration 7 $\rightarrow$ 1 2 3 / 4 5 6 / 0 7 8; Iteration 8 $\rightarrow$ 1 2 3 / 4 5 6 / 7 0 8, whose expansion generates the Goal State.

- 2) Total states generated: A full breadth-first search of this puzzle generates on the order of ~500 distinct states (branching factor 2–4 per state) before the goal is dequeued at depth 8. The exact count was verified computationally; the key point is that BFS's memory/state cost grows very quickly (roughly exponentially) with solution depth.
- 3) Complete solution path (8 moves): 1 3 6/5 0 2/4 7 8 → (Right) → 1 3 6/5 2 0/4 7 8 → (Up) → 1 3 0/5 2 6/4 7 8 → (Left) → 1 0 3/5 2 6/4 7 8 → (Down) → 1 2 3/5 0 6/4 7 8 → (Left) → 1 2 3/0 5 6/4 7 8 → (Down) → 1 2 3/4 5 6/0 7 8 → (Right) → 1 2 3/4 5 6/7 0 8 → (Right) → 1 2 3/4 5 6/7 8 0 (GOAL).
- 4) Why BFS always finds the shortest solution in an unweighted 8-puzzle: every move (sliding one tile into the blank) has the same cost (1 step). BFS explores the state space level by level in order of increasing depth from the initial state, so it never visits a state at depth $k+1$ before it has visited every state at depth k . Consequently, the very first time the goal state is reached, it must be via the minimum possible number of moves — a longer path to the goal simply cannot be discovered before a shorter one when all edge costs are equal.

(Rubric section retained from the original assessment sheet — omitted here for brevity in this answer key.)